Riley Beenders

ENGINEERING PROGRAMS | PRODUCT DEVELOPMENT | SYSTEMS IMPLEMENTATION

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PROFESSIONAL SUMMARY

Engineer who carries technical projects from a customer's need through scope, budget, sourcing, build, launch, and after-sales support, keeping engineers, vendors, and customers working from the same plan. Builds the tracking and reporting tools that give teams and management a live view of status and production, and explains technical decisions at the level of detail each audience needs.

CORE EXPERTISE

Project Management & Delivery: Requirements and scope definition with customers and external partners; proposals, scopes of work, and estimates; project budgeting and unit-cost tracking; coordinating schedules, sourcing, and deliverables across concurrent projects; risk analysis and mitigation; vendor communication and cost negotiation; product development and commercialization; Lean manufacturing

Tracking, Reporting & Systems: Microsoft Business Central ERP; custom ERP/CRM and production dashboard development; engineering documentation and BOM control; data logging and process analysis; Python; LabVIEW; Kubernetes and Docker; Git and GitHub; MATLAB

Engineering: Design for manufacturing and assembly (DFM/DFA); electrical layout and schematic design; GD&T; fixture and tooling design; R&D prototyping and validation; product safety and regulatory compliance; SolidWorks, Autodesk Inventor, Siemens NX, CATIA, Fusion 360

Infrastructure & Communication: UniFi multi-site network administration; VLAN segmentation; Windows/Linux systems administration; technical writing; communicating with technical and non-technical audiences; customer training; hiring and technical interviews; worldwide project collaboration

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA — January 2025 - Present

Lead, Research & Development Engineer

Proteor acquired Filament Innovations in January 2025, and the CTO role below continued as Lead R&D Engineer for the same product line, now Proteor Print.

- Owned the ICARUS-Lite from concept through commercialization against unit goals of 60 in 2025 and 100 in 2026, contributing to increased revenue: led its mechanical, electrical, and software design, cut assembly time by 80% compared to previous designs, and completed the product changes supporting conformity with the 2006/42/EC Machinery Directive.
- Managed the Coplay location's Microsoft Business Central ERP implementation in five weeks: two weeks evaluating the software, then three migrating more than 2,000 items and 150+ products into the live system, reconciling inventory counts on the floor, and syncing SKUs with Proteor's global locations.
- Estimate and track the budget of every engineering project, build its BOMs, and maintain each product's unit cost in Business Central; present project status, constraints, and budgets at Proteor's monthly USA review with leadership from its global locations.
- Aligned the division's software, IT infrastructure, and internal processes with Proteor's company-wide systems after the acquisition. Replaced a consumer mesh network that was assigning duplicate IP addresses with a centrally managed UniFi network, using VLANs to separate production, work, and IoT devices, which reduced IP conflicts and network downtime.
- Performed risk analysis and mitigations to EU and U.S. standards across ProteorPrint hardware and cloud products.
- Defined the machine requirements and integration for ProteorPrint OS with an external software partner (SimplyPrint), giving users remote operation, live video, AI-assisted failure alerts, and material, waste, and outcome metrics for analytics-driven operational decisions.
- Gave operators, the packing bench, and management live visibility into three filament extrusion lines with ExtrusionStudio, taking it from a LabVIEW proof of concept to a Python service that polls every line's laser gauge at 20 Hz, alarms when product leaves tolerance, and uses a laser-to-spool calibration step to log each reading onto the spool it is actually wound on. Every spool gets a serial, CSV record, certificate, and QR label that the packing bench approves or rejects against.
- Rolled ExtrusionStudio out across the plant on a fleet of Raspberry Pi kiosks (projector, per-line touch screens, packing bench, statistics TV) that check in with the server every 30 seconds and are updated, configured, and switched on and off with the work hours from one settings panel.
- Standardized POSEIDON and ICARUS on one linear-motor architecture with swappable mounts, reducing product-to-product complexity and simplifying service; validated 4-micron precision at 4 m/s with a 30 kg moving mass.
- Co-developed a custom hotend with an external manufacturing partner, improving reliability and reducing cost for the company's materials.

- Provide cross-functional technical support for the product line, from troubleshooting to after-sales service.
- Act as Hiring Manager for engineering roles, conducting technical interviews and owning final hiring decisions.

Filament Innovations | Whitehall, PA — May 2020 - January 2025

Across these roles, wrote customer proposals, scopes of work, and estimates, and began tracking project budgets.

Chief Technology Officer — January 2023 - January 2025

- Oversaw the product-development pipeline, internal IT infrastructure, and technology strategy while keeping hands-on responsibility for mechanical, electrical, firmware, and software systems, and gave the CEO and the company regular updates as designs and software iterated.
- Replaced separate manual workflows with a self-developed ERP/CRM that tracked orders, inventory, tasks, projects, and customer interactions in one system, generated BOMs dynamically, and linked design data to purchasing; designed and hosted it on a redundant Kubernetes cluster.
- Led the design and launch of Gen3 POSEIDON, the flagship's production redesign, combining a 1 m^3 build volume, filament-and-pellet dual extrusion, and braked Z drives that prevent uncontrolled motion during power loss.
- Researched high-speed linear motors that moved a 30 kg payload at 3 m/s with 5-micron accuracy and conducted early 6-axis robotic non-planar printing research.

Lead Electro-Mechanical Engineer — June 2022 - January 2023

- Removed a manufacturing bottleneck with independently testable electronics enclosures that installed in minutes, coordinating custom harnesses, cut panels, and hardware improvements with vendors.
- Integrated a multi-board CAN-bus control architecture that simplified wiring and debugging on larger, modular machines.
- Improved thermal and acoustic performance through enclosure redesign and component placement, including custom tooling with local fabricators for large-radius aluminum-panel bends.

Industrial Engineer — April 2021 - June 2022

- Delivered THE KRAKEN, a 1 x 2 x 1.5 m hybrid pellet-and-filament gantry machine for a client producing the world's largest 3D-printed statue, managing CAD, BOM, sourcing, fabrication, assembly, firmware tuning, customer support, and operator training.
- Designed and released Kratos, a machine engineered to print high-strength above-knee and below-knee prosthetic sockets.
- Developed next-generation ICARUS and POSEIDON models with higher-performance components and expanded motion safety through NEMA 34 motors, planetary gearboxes, and power-failure brakes.

Engineering Technician — May 2020 - April 2021

- Diagnosed mechanical and electrical problems during assembly and operation, improving early product reliability.
- Tuned, calibrated, and optimized machine parameters while documenting design limitations and improvement opportunities.

SELECTED PROJECTS

BulkSSH | 2026 | Electron, TypeScript | github.com/RileyBeenders/BulkSSH | Built to cut the testing time of the ExtrusionStudio rollout, where every change to the Pi image had to be tried on every Pi one SSH window at a time. Opens many devices in a grid of live terminals, mirrors keystrokes, and broadcasts a command with each device's output and exit code; destructive commands need confirming before they reach several devices. Released as an installer and a portable executable.

ADDITIONAL RELEVANT EXPERIENCE

Northampton Area School District - Computer / Network Technician | June 2016 - August 2019 | Built network infrastructure and deployed more than 750 wireless access points across six school buildings; accelerated deployment of more than 2,500 Chromebooks with an Arduino-based tool that automated setup-data entry, and coordinated final device distribution with school administrators.

RJB.Engineering - Owner & Engineering Consultant | 2015 - Present | Deliver project-based engineering support for automotive clients and operate a small online store for printer heat inserts and accessories.

Blue Mountain Ski Resort - Ski Lift Operator | November 2019 - January 2021 | Operated lifts and assisted thousands of guests with ticketing, weather, and slope information; performed motor and control safety checks and supported the maintenance team during lift breakdowns.

EDUCATION & CERTIFICATIONS

Penn State University | B.S., Electro-Mechanical Engineering — December 2024

MATLAB, Simulink, Simscape, and Machine Learning Onramps, MathWorks	2026
Lean Six Sigma Yellow Belt, Educate 360	2025
Lean Six Sigma White Belt, Educate 360	2025
Certificate in Engineering Design and Tools, Penn State University	2023
Mechanical Engineering Certificate, Lafayette College	2019